33 A beta particle has less mass than an alpha particle, but the speed of a beta particle emitted during radioactive decay is approximately 20 times greater than the speed of an alpha particle.

Which statement correctly compares properties of these two particles?

A Alpha particles are less penetrating because they have less kinetic energy than beta particles.

B Alpha particles are more penetrating because they have less kinetic energy than beta particles.

C Alpha particles are less ionising because they have more kinetic energy than beta particles.

D Alpha particles are more ionising because they have more kinetic energy than beta particles.